Maths Exploring for terminal solutions-13

## Corrected MCQs

1. **A.** By linearity, $E(Y) = 2E(X) - 1 = 5$. The constant is subtracted once.
2. **C.** The variance is multiplied by $(-3)^2 = 9$, and a translation does not change it. negative.
3. **B.** The sum has variance $n\sigma^2$, then division by $n$ multiplies variance by 1/ $n^2$.
4. **D.** The linearity of expectation does not require independence. additional conditions.
5. **C.** $E(X) = 1$, $E(X^2) = 2$, so $V(X) = 2 - 1^2 = 1$.
6. **A.** It's an increase in a gap to expectation, not an equality or a point bearing directly on $X$.
7. **D.** Solve 25/ $n \leq 0.05$ with $n > 0$ given $n \geq 500$. It is a sufficient size according to this bound.
8. **B.** A probability is always at most 1; the increase 2 does not give any improvement.
9. **C.** This is a convergence in probability, not a decrease in error at each observation.
10. **A.** The average is exactly $X$. Repeating the same variable does not create new observations independent.

**True/false.** False: the increase can be large. True: take the root of $a^2V(X)$. False: question 10, where variance does not decrease.

## Corrected applications and support

**Application 1.** Sum of probabilities 1. E=2; $E(X^2) = 11/2$, therefore $V = 3/2$ and $\sigma = \sqrt{6}/2$.

**Application 2.** $E(Y) = 4$, $V(Y) = 27/2$ ; $E(Z) = -2$, $V(Z) = 3/2$. The square of the deviations to the average does not change under sign inversion.

**Implementation 3.** Independence: variance 1 $/(4n)$. Identical copies: the average is X, so variance 1/4 whatever n. Repeating the same measurement does not create an independent sample.

**Implementation 4.** The bound is 9/ $(0{,}25n) = 36/n$. Require $36/n \leq 0.01$ given $n \geq 3600$.

**Application 5.** The event is $X \leq 2$ or $X \geq 8$. By symmetry, its probability is 2 $(1 + 10 + 45)/2^{10} = 7/64 \simeq 0{,}109375$. The bound is $(1/40)/(9/100) = 5/18 \simeq 0{,}2778$, so it's correct But less precise.

**Implementation 6.** Theoretical standard deviations are about 0 ,1581, 0,05 and 0,01581. Simulated outputs fluctuation: a set of means is compared, not the successive values of a single average. Empirical proportions are not demonstrations. The reproducible results file gives the values obtained with the seed retained.

**Implementation 7.** Results 0.1 give frequency 1 /2 ; add 1 gives 2 /3, further from 1 /2. For the probability of a deviation at least equal to that accuracy tends to 0 when n This does not impose a monotonous rapprochement on each achievement.

**Reading the guarantees**

Surround the strict sign of the central event and the broad sign of its complement. Compare a computable probability and a universal limit. Do not present the simulation as evidence of the law of large numbers.

## Conduct and correction

**Benchmarks for the teacher**

The DS5 and the second white precedes S73: do not ask for Bienaymé-Chebychev nor law
for variance, make surround the hypothesis of independence.
concentration, distinguishing exact probability and increase, then sufficient rank and minimum rank
A simulation is not evidence and an observed average does not approach
Of course, every time you try again.

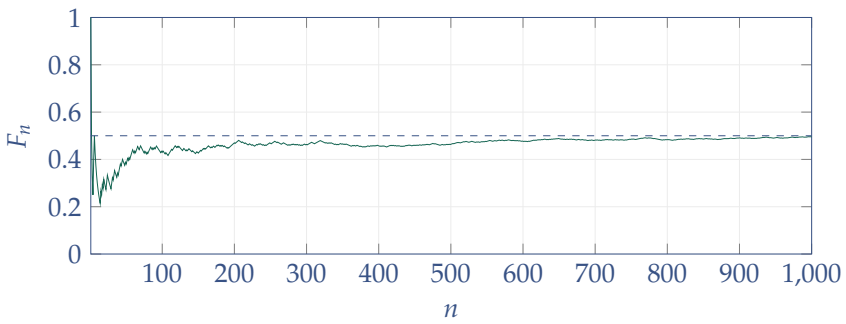
### Corrected activity

1.

```
From random import Random

def trajectories_piece(N=1000, seed=2026):
    rng = Random(grain)
    batteries, frequency = 0, []
    for n in rows (1, N+1):
        batteries += rng.random() < 0.5
        frequency.append(cells/n)
    return frequency
```

Trace points $(n, F_n)$ with the list obtained. After two tests giving a battery and one face, $F_2 = \frac{1}{2}$. Whatever the third result, $F_3$ value $\frac{1}{3}$ or $\frac{2}{3}$ : difference at $\frac{1}{2}$ Stabilisation increases. is not a monotonous decrease of the deviation on each trajectory.




A simulated trajectory, seed 2026; the right is $y = \frac{1}{2}$.

2. Note: $B_i$ I'm the battery-indicator. $F_n = \frac{B_1+\cdots+B_n}{n}$. Each $B_i$ has expectation $\frac{1}{2}$ and variance $\frac{1}{4}$. By linearity, $E(F_n) = \frac{1}{2}$ ; by independence and change of scale, $V(F_n) = \frac{n(\frac{1}{4})}{n^2} = \frac{1}{4n}$.

3. Bienaymé-Chebychev gives $P(|F_{1000} - \frac{1}{2}| \geq 0{,}05) \leq \frac{1}{4\times 1000\times 0{,}05^2} = 0{,}1$. It is a bound higher, not the exact probability.

4. To ensure a complementary probability of up to 0 ,01, it is enough to $\frac{1}{4n\times 0{,}05^2} \leq 0{,}01$. This is equivalent to to $n \geq 10000$. $P(|F_n - \frac{1}{2}| < 0{,}05) \geq 0{,}99$.

5. The simulation illustrates a finished realization; other seeds give other trajectories. establishes a guarantee of probability for any sample meeting the assumptions of independence and common law.

**E1.** The affine transformation gives $E(4X-1) = 4E(X) - 1 = 11$ and $V(4X-1) = 4^2 V(X) = 32$. The standard deviation is $\sqrt{32} = 4\sqrt{2}$ ; the lag $-1$ does not change dispersion.

**E2.** By linearity, $E(2X-Y) = 2E(X) - E(Y) = 4 - 5 = -1$. The independence of $X, Y$ allows add the variances of 2 $X$ and $-Y$ : $V(2X-Y) = 4V(X) + V(Y) = 4 + 4 = 8$.

**E3.** The average of 64 independent variables of the same law retains expectation 10. value $\frac{4^2}{64} = \frac{1}{4}$, so its standard deviation is $\frac{1}{2}$.

**E4.** Bienaymé-Chebychev with $\delta = 6$ given $P(|X-100| \geq 6) \leq \frac{V(X)}{6^2} = \frac{9}{36} = \frac{1}{4}$. This bound does not does not give an exact equality or value.

**E5.** We have $P(|M_n - \mu| \geq 0{,}2) \leq \frac{4}{n \times 0{,}2^2} = \frac{100}{n}$. For this bound to be not more than 0 ,05, it is enough to $n \geq 2000$. The complement then has a probability of at least 0.95.

### Corrected from DM14

1. Note: $B_i$ independent indicators of success. $F_n = (\frac{1}{n}) \sum_{i=1}^{n} B_i$, $E(B_i) = p$, $V(B_i) = p(1-p)$. So

$$E(F_n) = \frac{1}{n}\sum_{i=1}^{n} p = p, \qquad V(F_n) = \frac{1}{n^2}\sum_{i=1}^{n} p(1-p) = \frac{p(1-p)}{n}.$$

2. $p(1-p) = p - p^2 = \frac{1}{4} - (p - \frac{1}{2})^2 \leq \frac{1}{4}$. Equality is achieved for $p = \frac{1}{2}$.

3. For every $p \in [0,1]$, $P(|F_n - p| \geq 0{,}02) \leq \frac{1}{4n \times 0{,}02^2}$. It is sufficient that this bound is not more than 0,05, i.e. $n \geq 12500$. By addition, this size guarantees the required probability, uniformly in $p$.

4. On $[0,0{,}1]$, function $p \mapsto p(1-p)$ is increasing as its derivative 1 $- 2p$ is positive. $p(1-p) \leq 0{,}1 \times 0{,}9 = 0{,}09$. It is now sufficient to $n \geq \frac{0{,}09}{0{,}05 \times 0{,}02^2} = 4500$. These sizes are sufficient within the meaning of the mark-up, not necessarily minimum for the exact probability.

5. If all tests copy the same variable $B$, then $F_n = \frac{B + \cdots + B}{n} = B$. So $V(F_n) = p(1-p)$, without division by $n$. The addition of variances was permitted only through independence; The hypothesis disappears here. $< p < 1$, increase the number of copies does not bring any new information.

### Corrected work in the manual or an anal

$M_N$ is the average number of universal donors per sample of 100 people. Linearity gives $E(M_N) = 7.14$; independence gives $V(M_N) = \frac{6{,}63}{N}$.

$$P(|M_N - 7{,}14| \geq 0{,}14) \leq \frac{6{,}63}{0{,}14^2 N}.$$

To make this bound equal to or less than 0.05, just take

$$N \geq \left\lceil \frac{6{,}63}{0{,}05 \times 0{,}14^2} \right\rceil = 6766.$$

For additional information, $P(7 < M_N < 7{,}28) \geq 0{,}95$.